Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/20/2018
PI Name:	Espinosa, Araceli Ph.D.		
Project Title:	The Impact of Real Microgravity on the Proliferation of Human Neural Stem Cells and derived-Oligodendrocytes		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	05/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/12/19) NOTE: Extended to 11/30/2019 per F. Hernandez/ARC (Ed., 10/24/18) NOTE: Extended to 10/31/2018 per F. Hernandez/ARC (Ed., 11/2/17)		
Key Personnel Changes/Previous PI:	September 2015 report: No changes have occurred.		
COI Name (Institution):			
Grant/Contract No.:	NNX15AB43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent data from NASA shows that astronauts present vision, physiological, and anatomical changes as well as intracranial pressure (ICP) and visual impairment intracranial pressure (VIIP) after long-duration space missions. These changes are unlike any clinical entity previously seen. Yet, the cause(s) of chronically elevated ICP/VIIP and potential consequences are unknown. The main goal of this International Space Station (ISS) spaceflight proposal is to obtain reference data on the effects of real microgravity (0G) to validate and expand our recent data obtained using simulated microgravity (sim-μG). We want to assess the impact of real microgravity on the cell cycle of human neural stem cells and derived oligodendrocytes (OL) that are the myelin forming cells in the central nervous system. We now have evidence that sim-μG increases the number of neural progenitors and OL progenitor (OLP) via cell proliferation. Here we postulate that microgravity-associated VIIP and altered ophthalmic anatomy (papilledema, choroidal folds, etc.) as well as a potential microgravity-associated ICP without visual impairment might be at least in part due to increased cell proliferation in the brain. Aim I. To test the hypothesis that neural stem cells (NSC) and OL progenitors exposed to real microgravity will proliferate faster and will give rise to more progenies than cells grown in 1G. Aim II. To test the hypothesis that in real microgravity NSC and OL produce and secrete one or more molecules that may act as regulators of cell proliferation. Significance. Neural cells possess intrinsic plasticity that can be modulated by the environment and their response to real microgravity may confirm and even reveal new properties of these cells triggered by it. We also propose to examine molecules secreted by these cells while in space in order to identify potentially unique regulators of cell proliferation. It is known that secretome proteins play a key role in cell signaling and cell migration. These molecules may also hold a potential key for the modulation of normal cell proliferation for future therapeutic interventions for the central nervous system (CNS). Approach. Human NSC and OLP will be plated using our proprietary specific culture media in self-contained hardware designed for 0G (Astrium). Control cells will be maintained in 1G. Ground based experiments. We will analyze the effects of real microgravity on NSC and OL cell cycle via immunocytochemistry. After landing and recovery of the samples, the conditioned culture medium will be collected, maintained at 4°C, and transported to our laboratory to further characterize the secretome of both cell types maintained in real microgravity. Our stem cell culture medium as well as the oligodendrocyte specification medium are both serum-free and contain few proteins, properties that will facilitate the secretome analysis. The characterization of the secretome will include identification of proteins and small molecules that may be produced and secreted in real microgravity only. Innovation. This is the first study on the impact of real microgravity on human neural cells. Our data will provide valuable information to better understand the gravitational epigenetic modulation of human neural cells. Benefits for Humanity. Confirming that real microgravity increases cell proliferation will move us forward toward a better understanding on the causes of ICP and VIIP in astronauts. Moreover, since long-term missions in space are imminent, the identification of molecules that increase cell proliferation might lead to the design of potential countermeasures to prevent or alleviate the negative effects observed in astronauts exposed to long-term missions in space. Myelin is responsible for the rapid conduction of electrochemical messages to and from the brain. Since there is no cure for myelin disorders our data might also lead to novel therapeutic modalities for myelin repair.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To date, there is no cure for developmental disabilities, neurodegenerative disorders, myelin loss, or cancer. We expect that our work described in the current proposal will generate cures that are of paramount importance in two different but complementary fields: 1) Drug discovery: The first product will be the first molecule that would have been identified, isolated, and tested to be responsible for increased cell proliferation. This will open the door for a second product (or series of products) under this category that would be an antibody or similar molecule(s) that would void/neutralize the effects of the first molecule on cell proliferation. This second product will be further studied as an anti-cancer agent via prevention of cell proliferation. If there is more than one relevant secreted molecule a similar process would be followed. Once we have data showing that we have interesting molecules, we will submit an invention report to the Office of Intellectual Property at UCLA. This office supports scientists by contacting potentially interested companies and helping disseminate information on technologies available for licensing. In some cases UCLA technologies are commercialized by large companies. Alternatively, some entrepreneurs-inventors form successful startups based on their own UCLA technology. 2) Generation of cells for cell replacement therapies: The presence of functional myelin is essential for all our normal daily activities. Our overall goal is to investigate the prospects of cell replacement therapies for patients that require normal OL and myelin to restore proper CNS function. We contemplate to be able to apply this approach to propagate OLPs also known as (OPCs) and make them available to treat the spectrum of myelin disorders, from white matter damage in the premature neonate and infants suffering of autism or maternal prenatal infection, to the patient with leukodystrophies, multiple sclerosis, psychosis/schizophrenia, and trauma. We have previously demonstrated that OLPs are cells that, once grafted, migrate, integrate, and interact with the host CNS. Validating the functionality of OLPs exposed to real μG will open a new era on the ways we conceive cell replacement therapies and CNS repair. Moreover, trauma to the brain and spinal cord involves myelin loss and glial and neuronal cell death; thus, data generated by the current study will provide insights on producing cells to be transplanted into the CNS of individuals suffering from trauma. Prior to commercializing the generation of cell populations in microgravity, it is essential to confirm that microgravity-exposed cells are fully functional and safe; by biochemical and pre-clinical neural transplantation proposed in this grant application we will generate the necessary evidence. Once this step will have been accomplished, we'll contact specific companies that might provide us with the revenue necessary to fund the scale-up process to generate human induced pluripotent stem cell (hiPS) derived oligodendrocytes progenitors (OLP). By then we hope to have created strong partnership with the industry in order to be able to generate cells to overcome neural degeneration and developmental disabilities.

The Impact of Real Microgravity on Neural Stem Cells and Oligodendrocyte Progenitors: Ground based experiments: Preflight Assessments
Project Objectives: The goals of this project include determining proliferation rates of human neural stem cells (NSCs) and oligodendrocyte progenitor cells (OLPs also known as OPCs), both in microgravity. To perform this experiment, cells will fly to and remain on the ISS for a 30-day term. These cells have several growth constraints including atmospheric composition (i.e., CO2 and O2 concentrations) and temperature. We shall also examine the secretome of these progenitors after exposure to real microgravity. Previously, using OLs the myelin forming cells in the central nervous system CNS), OSM (oligodendrocyte specification medium) a chemically defined medium designed by us (Espinosa et al., 2009; 2016a), and a 3D-Clinostat we subjected these cells to sim-0G and found enhanced and sustained proliferation, with a concomitant shortening of the cell cycle. We have also observed some other significant changes in biological processes, including oxygen consumption rates, that are enhanced in OLPs exposed to sim-0G (for details see previous report and Espinosa et al., 2013). Moreover, these cells also secrete numerous fatty acids and more complex lipids in just three days while this was not observed in 1G sister cultures (Espinosa et al., 2016b).

Year 4 Report

Experimental Verification Tests (EVT): This year we performed the Experimental Verification Tests (EVT) that are the last step to be accomplished before getting authorization for flight implementation. The EVT activities were started at the end of December 2017 and continued until January 13, 2018. Data was collected from cells kept up to 51 days in the spaceflight containers in the specific conditions that our neural cells will experience during ascent, on board of the International Space Station and during descent/re-entry and splash-down.

We used various methods to ascertain the viability and performance of both cell types, neural stem cells (NSC) and oligodendrocytes (OLs) the cells that make and maintain myelin in the central nervous system (CNS). These tests were performed for both cell types and using both types of hardware: passive and automated. Light microscopy was used to determine cell morphology and events such as proliferation and maturation of the cells prior to starting the experiment and upon conclusion of the “travel and incubation times.”

We have previously reported that OLs consume more energy in simulated microgravity when compared to cells maintained in 1G, i.e., terrestrial gravity.

Energetic status of our cells: We selected the SeaHorse Bioscience assay because it is an ideal quantitative indicator of the metabolic state of the cells; for example, if the oxygen consumption rate (OCR) is deficient the cells might not be in optimal conditions.

The use of antibodies to identify specific cell marker expression and typical stage-specific markers together with cell morphology are very helpful techniques to establish the developmental stage of the cells. We examined the cells after each time point from 45 to 55 days unattended using fluorescent microscopy.

We also ascertained the amount of protein contained in the culture medium alone and the one contained proteins secreted by the cells “secretome” in order to ensure that we would have enough protein per container for the proteomics study.

Results

• Human oligodendrocyte progenitors HuOLPs survive well unattended for up to 51 days and recover well in our optimized OLs culture medium.

• NSCs survived well 51 days unattended and recovered fast upon being back to 37degreesC in fresh medium as assessed by the Energetics, morphological and fluorescent expression of cell markers.

• The secretome of both passive and active units contained enough protein for our study.

Task Progress:

• In order to protect the samples prior to re-entry to the atmosphere, we have to arrest the growth of the cells. We tested fixative and RNAlater in order to elucidate if these substances are compatible with immunofluorescent detection of surface and internal cell markers.

• RNAlater is compatible with protein detection and quantification in the secretome of our cells.

Understanding CNS changes elicited by exposure to real microgravity may lead us to address disturbances such as intracranial hypertension experienced by astronauts in space and/or upon return to Earth.

In summary

Experiments were finished in a successful manner. On these bases “success criteria for specific parameters were established.”

In parallel we worked on the “EXPERIMENT REQUIREMENTS DOCUMENT” that is necessary to provide details about the requirements for our cells while at the Cape, during ascent, descent, and while in the ISS.

We are ready since March 2018, as our cells were supposed to fly in SpaceX-14 on April 2, 2018. However, a malfunction of the STaARS-1 Platform prompted re-scheduling the flight of our cells to November 29, 2018 in Space X-16.

Conferences

Invited Speaker at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, October 2017. “Human oligodendrocytes in simulated microgravity generate a unique secretome metabolic profile.” LS Systems II: Gravity Sensing, Vision and the Brain. Seattle, Washington.

Invited Speaker at the CSA/ESA/JAXA/NASA ISS Increments 55/56 Science Symposium, January 2018. “The Impact of Real Microgravity on the Proliferation of Human Neural Stem Cells and Derived-Oligodendrocytes.”

References

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Bibliography Type:	Description: (Last Updated: 02/07/2024)
Abstracts for Journals and Proceedings	Nguyen K, Espinosa-Jeffrey A. "Human Oigodendrocytes in Sim-0G but Not in 1G Secrete Numerous Lipids in Just Three Days." Oral presentation in scientifc event F2.4, Lecture Room SR 4 (PCC). Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. , Jul-2018
Abstracts for Journals and Proceedings	Espinosa-Jeffrey A. "Human oligodendrocytes in simulated microgravity generate a unique secretome metabolic profile." LS Systems II: Gravity Sensing, Vision and the Brain. Seattle, Washington. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Espinosa-Jeffrey A. "The Impact of Real Microgravity on the Proliferation of Human Neural Stem Cells and Derived-Oligodendrocytes." CSA/ESA/JAXA/NASA ISS Increments 55/56 Science Symposium. Video-conference. Invited speaker. CSA/ESA/JAXA/NASA ISS Increments 55/56 Science Symposium, Seattle, Washington, January 2018. , Jan-2018
Articles in Peer-reviewed Journals	Espinosa-Jeffrey A. "Physiologically-relevant cellular models for biomarkers and drug discovery." Annals of Stem Cell Research & Therapy. 2018 Apr 26;2(2):Article 1014. http://www.remedypublications.com/annals-of-stem-cell-research-and-therapy/articles/pdfs_folder/ascrt-v2-id1014.pdf , Apr-2018